

Computer Architecture (CS F342)
Semester-I, 2022-23
Recheck for Midsem Answer
Department of Computer Science and Information Systems (CSIS)
BITS-Pilani, K K Birla Goa Campus, Goa, India.

Due Date for sending recheck request: Nov 18, 2022: 1:00 PM

Instructions:

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

1. (a) Decoder complexity will increase from $6:2^6$ to $12:2^{12}$
(b) No next state.
(c) BNE, J. Instruction alignment problem.
(d) or **\$t2**, \$t4, \$t5 and addi \$t1, **\$t1**, 50

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

Recheck request link: https://docs.google.com/forms/d/e/1FAIpQLSf8XDn0xt5L-KRDRdb_i8-FHGI3Urjs25shKE2paVaE5kUfJw/viewform

2. (a) R-type instruction. \$t0 = \$t0 funct \$t0.
(b) Reg. fields will be increased from 5 bit to 7 bit. Opcode fields will be increased from 6 bit to 8 bit. For I-type instruction, Imm field can be reduced.
(c) [2FFF E000, 3001 FFFC] or [3002 0000, 2FFE 0004]
(d) 26 or 21
(e) $M = 3 * (\&D[0])$

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

Recheck request link:

<https://docs.google.com/forms/d/e/1FAIpQLScjoV8y6wu6lqlbutIRmuZtcNcfc054uPUPVCrr0O6PiwvgDg/viewform>

3. NA

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

Recheck request link: <https://forms.gle/9Xjg1C7pevr8HQFY7>

- 4.

- (a) 725 ps.
(b) Slower. Execution time: 1000.5 ps.
(c) Number of lw & sw

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

Recheck request link:

https://docs.google.com/forms/d/e/1FAIpQLSefnXy5zvTXFCNt80sxs27jloScIlehl_meO1KIHjwNLCymiQ/viewform

5. NA

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

Recheck request link:

https://docs.google.com/forms/d/19Yt1eFJfDrSCOiLlnLxCpRwTjxXLaOnCd7MMxCJqTaw/viewform?edit_request_ed=true

- 6.

- (a) Speedup = 1.11
(b) 0.2
(c) 1)

```
add $s3, $s1, $s0
NOP
NOP
lw  $s3, 4 ($s3)
lw  $s1, 0 ($s4)
NOP
or  $s2, $s3, $s2
NOP
NOP
sw  $s2, 0 ($s3)
```

2) No change

Unnecessary recheck request will deduct -5 marks from the total. One may get negative marks also.

Recheck request link: <https://forms.gle/26Whq21w3CizAryF6>